Weather Classification Multivariate Analysis Project Report

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I. Introduction

This project analyzes a weather dataset of 13,200 observations with 7 numerical variables and one categorical class variable. The aim is to examine how weather-related variables distinguish between weather categories and to evaluate the effect of outliers on classification performance. Our methodology identifies outliers using the BACON algorithm, applied to the entire dataset and to each class separately and discriminant analysis (FLDA and Multinomial) is performed both before and after outlier removal. The central research question is: How do outliers affect multivariate classification performance, and how can robust methods improve results using FLDA and Multinomial regression?

II. Data Description

The dataset (Table I) used in this analysis is obtained from Kaggle and consists of weather-related observations. For the purpose of this study, 7 numerical variables and one class variable representing the weather type are selected. These variables are used to explore the relationships between different weather conditions and to support the subsequent outlier detection and classification analyses.

III. BACON Algorithm Results

III.A. ***Whole dataset***

1. Subsets of MV BACON

- (subset no. 1): 28 of 13200 (0.21 %)
- (subset no. 2): 9219 of 13200 (69.84 %)
- (subset no. 3): 11040 of 13200 (83.64 %)
- (subset no. 4): 11523 of 13200 (87.3 %)
- (subset no. 5): 11904 of 13200 (90.18 %)
- (subset no. 6): 12069 of 13200 (91.43 %)
- (subset no. 7): 12110 of 13200 (91.74 %)
- (subset no. 8): 12118 of 13200 (91.8 %)
- (subset no. 9): 12120 of 13200 (91.82 %)
- (subset no. 10): 12121 of 13200 (91.83 %)
- MV-BACON (subset no. 11): 12121 of 13200 (91.83 %)

Number of outliers: 1079

Outlier row indices: 11, 39, 59, 73, 86, 94, 95, 110, 114 . . .

The BACON algorithm constructs basic subsets of observations iteratively. It starts with an initial subset of size $m = 28$, consisting of observations with the smallest distances. At each step, robust Mahalanobis distances are computed using the mean and covariance of the current subset, assumed to be largely free of outliers. A new subset is formed by including observations with distances below the critical value c_α, while others are flagged as potential outliers. The process stops at the 11th iteration, as the 10th and 11th subsets are identical, indicating convergence to an outlier-free subset which includes 12121 observations and the remaining 1079 observations are outliers.

2. Bacon Limit

BACON limit: 6.1275

The BACON limit ($c_\alpha = 6.1275$) is the cutoff used to distinguish regular observations (basic subset) from potential outliers (non-basic subset). Observations with distances above this threshold are flagged as outliers, while those below are considered part of the main dataset.

In the BACON distance plot (Fig I), the horizontal red line represents this limit: points above it are outliers, and those below are normal observations.

The Q–Q plot (Fig I) shows strong deviation from the theoretical chi-square distribution. While lower values roughly follow the reference line, the upper tail deviates substantially, indicating extreme observations and significant outliers.

Note that:

X-axis: Chi-square quantiles (expected under multivariate normality)

Y-axis: Squared BACON distances (BD)

Blue line: reference line (ideal case under the assumption)

The index plot of Mahalanobis distances (Fig I) shows many observations exceeding the cutoff, suggesting numerous outliers. Its Q–Q plot (Fig I) also deviates strongly from the chi-square line, indicating violation of multivariate normality. However, these results may be misleading since Mahalanobis distance is non-robust, relying on the mean and covariance, which are highly sensitive to outliers. In contrast, BACON uses a robust subset, leading to a smaller and more meaningful set

of detected outliers that better reflects the data structure.

III.B. Each Class

In this section, the bacon algorithm is applied to 4 separate datasets for the four classes to assess the quality of outliers detection using Bacon when it's applied on each class separately.

Bacon Output:

1) Numerical

BACON Outlier Detection (By Class):

The BACON algorithm is applied separately to each class (Cloudy, Rainy, Snowy, and Sunny) in order to detect outliers within each group.

Cloudy (n = 3300)

Number of outliers: 720, BACON limit: 5.8742. The algorithm converges after testing different subsets and identifies 720 observations as outliers.

Rainy (n = 3300)

Number of outliers: 334, BACON limit: 5.8742. The algorithm tests multiple subsets and stabilizes after several iterations, resulting in 334 outliers.

Snowy (n = 3300)

Number of outliers: 629, BACON limit: 5.8742. The BACON procedure identifies 629 outliers after convergence of the iterative subset selection.

Sunny (n = 3300)

Number of outliers: 765, BACON limit: 5.8742.

Overall, the algorithm tests different subsets until it reaches a stable solution within each class.

2) Graphical:

This graphical output (Fig V) shows the Mahalanobis Distance (MD) index plots for each class as a non-robust method for outlier detection. In these plots, there is very little clear separation between the outliers and non-outliers. In contrast, the BACON index plots show a much more robust, well-defined line separating the outlier and non-outlier data. The standard Mahalanobis

results can be misleading because the method is not robust against extreme values.

We shall note that The identify() function is used in the Rscript file to interactively label individual outlier observations on the BACON distance index plot. By clicking on flagged points (those above the red dashed BACON limit), their row indices are displayed directly on the plot, allowing precise identification of which observations are considered outliers.

IV. FLDA Results

Checking the histograms of variables (Fig II) we found that not all variables are marginally normal, hence the data is not multivariable normal data.

Pairwise Scatter Plot (Fig VI): The pairwise scatterplot reveals that the relationships between variables are largely non-linear and not smooth elliptical clouds. This pattern indicates that the joint distributions are far from bivariate normal, which is consistent with the earlier finding that the data does not follow a multivariate normal distribution, which is an underlying assumption of FLDA.

*IV.A. **With Outliers***

IV.A.i. FLDA Internal Validation

1. A confusion matrix showing the classification performance across all weather categories.
2. A vector of predicted class labels assigned to each observation.
3. Posterior probabilities representing the likelihood of each observation belonging to each class.

1) **Confusion Matrix (Table II)**

   Error Rate = 14.42424%

   There are 2681 observations correctly classified as cloudy out of 3300, 2796 correctly classified as rainy of out 3300, 3049 correctly classified as snowy out of 3300, and 2770 correctly classsified as sunny out of 3300 using Fisher Linear Discriminant function. The error rate of classification is 14%. This means the summation of the incorrectly classified observations in each row (off diagonal entries) divided by the total number of observations is 0.14. Hence

14% of the observations are misclassified.

$$\text{Error rate} = \frac{324 + 97 + 198 + 158 + 252 + 94 + 98 + 39 + 114 + 214 + 233 + 83}{13200} \times 100 = 14.4$$

This indicates relatively poor separation between the classes.

2) **Class:**

These are the assigned classes to each observation. This assignment depends on the next output which is posterior (assigns each observation to the highest porbability class from posterior)

```
Rainy Cloudy Sunny Rainy Rainy
```

3) **Posterior (Table III)**

Porbabilities of each observation being in a specific class.

Note that the class of highest probability is the one FLDA assigns in the output class.

IV.A.ii. FLDA External Validation

We chose 20% (2640 observations) of the data to be testing data and trained the model on the rest of 80%

Confusion Matrix (Table V)

Error Rate = 15.03788 %

$$\text{Error Rate} = \frac{66 + 24 + 36 + 32 + 58 + 24 + 22 + 6 + 19 + 46 + 44 + 20}{2640} = 0.1503$$

The misclassified is the off-diagonal entries and the correct ones are the diagonal elements.

IV.A.iii. FLDA2

FLDA2 is performed using two numerical variables: Temperature and Humidity, and is restricted to two classes (Snowy and Sunny). The three outputs:

1) **Confusion Matrix (Table IV)**

Error Rate = 6.015152%

Similar to the case of FLDA, there are 3076 out of 3300 observations correctly classified as snowy and 3127 out of 3300 correctly classified as sunny. The error rate of 6.01% indicates the sum of the incorrect classifications (224 + 173) divided by the total number 6600 suggesting a better classification than the case of multivariable FLDA since we use only two variables and two classes (Sunny and Snowy) which are well separated. Moreover, the two variables (Temprature and Humidity) are highly discriminative features meaning that the model can clearly distinguish between the observations according to those two variables with low error rate.

2) **The graph (Fig III):-**

The graph indicates the line of fisher discriminant function (blue line), where observations on the right of the blue line are classified as one group and the observations to the left of it are classified as the other group. The green line joins the centroids of each class.

The class corresponding to the black points is most likely the Sunny class, while the pink points correspond to the Snowy class. This interpretation is supported by the distribution of the observations, as the region associated with the black points shows higher temperature and lower humidity, which is consistent with sunny weather conditions. In contrast, the pink cluster is located in a region of lower temperature and higher humidity, which aligns with snowy weather characteristics.

*IV.B. **Without Outliers***

IV.B.i. FLDA Internal Validation

The FLDA method provides three main outputs for internal validation:

1. A confusion matrix showing the classification performance across all weather categories.

2. A vector of predicted class labels assigned to each observation.

3. Posterior probabilities representing the likelihood of each observation belonging to each class.

1. Confusion Matrix (Table VI)

The overall error rate is computed as:

$$\text{Error Rate} = \frac{12 + 3 + 7 + 85 + 33}{13201} = 0.01302 \approx 1.30\%$$

Overall, the model achieves a very low misclassification rate, indicating strong classification performance.

The model correctly classifies 2565 observations as Cloudy, 2874 as Rainy, 2671 as Snowy, and 2502 as Sunny out of 3300 observations per class.

2. Predicted Classes

The predicted classes for the first 15 observations are:

```
Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy
Cloudy Cloudy Cloudy Cloudy Cloudy
```

3. Posterior Probabilities (Table VII) Porbabilities of each observation being in a specific class. Note that the class of highest probability is the one FLDA assigns in the output class.

IV.B.ii. FLDA External Validation

The FLDA method provides three main outputs for external validation:

1. A confusion matrix showing the classification performance across all weather categories.

2. A vector of predicted class labels assigned to each observation.

3. Posterior probabilities representing the likelihood of each observation belonging to each class.

1. Confusion Matrix (Table VIII)

The overall error rate is computed as:

$$\text{Error Rate} = \frac{2 + 2 + 2 + 3 + 17}{2151} = 0.0120874 \approx 1.21\%$$

Overall, the model shows a very low misclassification rate, indicating strong generalization performance on the external dataset.

2. Predicted Classes

The predicted classes for the first 15 observations are:

```
Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy Cloudy
Cloudy Cloudy Cloudy Cloudy Cloudy
```

3. Posterior Probabilities (Table IX)

Porbabilities of each observation of the test data being in a specific class. Note that the class of highest probability is the one FLDA assigns in the output class.

IV.B.iii. FLDA2

FLDA2 is performed using two numerical variables: Temperature and Humidity, and is restricted to two classes (Snowy and Sunny).

Classification Results (Table X)

The overall error rate is computed as:

$$\text{Error Rate} = \frac{1}{5206} = 0.0001920861 \approx 0.0192\%$$

The FLDA2 error rate is notably lower than the multivariate FLDA internal error of 1.30%, consistent with the with-outliers case. This superior performance is attributed to the focus on two well-separated classes (Sunny and Snowy) and two highly discriminative features (Temperature and Humidity), which together provide a clear boundary between observations and allow the model to classify them with minimal error.

The graph (Fig IV) shows the line of the Fisher discriminant function (blue line), where observations on the right side are classified as one group and those on the left side are classified as the other group. The green line connects the centroids of each class. After removing outliers, the clusters become more concentrated. This reduces the spread of the data and leads to a clearer separation between the two groups, with less overlap.

Overall, the FLDA2 model achieves an extremely low error rate, indicating near-perfect classification performance using the selected two variables and two classes.

Comparison:-

The FLDA model generalizes well to unseen data, as the external validation error closely mirrors the internal validation error in both cases (with and without outliers). Removing outliers improves performance: the internal error drops from 14.42% to 1.30%, and the external error drops from 15.03% to 1.27% in case of multivariable FLDA model and also FLDA2 model (6.015152 % vs 0.0192%), confirming that outliers have a strong negative impact on FLDA classification accuracy. Since FLDA depends on class means and covariance matrices which are non-robust i.e. they are affected with outliers.

V. Multinomial Results

V.A. ***With outliers***

In this section, the Multionmial classification method is applied to the dataset with the outliers

V.A.i. Multinomial Internal Validation

1. Confusion Matrix (Table XI)

Error Rate = 0%

Error rate is 0% which means the model performs perfectly since it under estimates the true miss classification error. This perfect error is likely due to the presence of extreme observations, which can affect the class separation during the model fitting. This may lead to a perfect or near-perfect classification.

V.A.ii. Multinomial External Validation

1. Confusion Matrix (Table XII)

Error Rate = 13.40909 %

$$\text{Error Rate} = \frac{57 + 26 + 42 + 28 + 32 + 25 + 20 + 5 + 19 + 42 + 39 + 19}{2640} = 0.13409$$

In the case of the Multinomial model, we achieve a relatively high external misclassification

rate than the internal error, which is an expected output since the external error is, in general, more than the internal. This indicates poor classification performance on unseen data in general. It also indicates that the presence of outliers increases the error of external validation, as we will see in the external validation of the model on data without outliers yields less error (0.18596% vs 13.4%), suggesting that in multinomial outliers make classification of unseen data more difficult.

V.B. ***Without outliers***

V.B.i. Multinomial Internal Validation

In this section, the Multinomial classification method is applied to the dataset without outliers.

The output of the Multinomial model is:

1. Confusion Matrix (Table XIII)

The internal error rate is computed as:

$$\text{Error Rate} = \frac{6+3}{\text{Total observations}} \times 100 = 0.08370536\%$$

Overall, the Multinomial model achieves a very low internal misclassification rate, indicating strong classification performance on the dataset without outliers. Moreover, The increase in internal validation error observed in the multinomial model after removing the outliers is because the model is trained on a cleaner dataset, which reduces this effect and gives a more realistic representation of the data structure. As a result, a small number of observations become misclassified, leading to a slightly higher internal error rate of 0.0837%.

V.B.ii. Multinomial External Validation

The Multinomial classification method is also evaluated on the external dataset.

1. Confusion Matrix (Table XIV)

The external error rate is computed as:

$$\text{Error Rate} = \frac{2+1+1}{2151} \times 100 = 0.18596\%$$

Overall, the Multinomial model demonstrates strong generalization performance on the external

dataset without outliers, with a very low misclassification rate compared to the case of presence of oultiers (13.4%) as we mentioned in our analysis of the data with outliers.

VI. Comparison of FLDA and Multinomial

In conclusion, the results show a clear effect of outliers on the performance of both FLDA and Multinomial models.

For the FLDA model with outliers, the error rates are quite high, with 14.42% for internal validation and 15.03% for external validation. After removing the outliers, the error rates drop significantly to 1.30% (internal) and 1.21% (external). This shows that FLDA is strongly affected by outliers.

For the Multinomial model, the internal error rate without outliers is 0.0837%, and the external error rate is 0.18596%. When outliers are included, the error rates increase, especially in the external validation (13.4%), while the internal error rate remains low (0%).

Overall, the dataset without outliers gives better results for both models as the errors decrease after removing them. FLDA improves a lot after removing outliers, while Multinomial performs more consistently and generally gives lower error rates.

In general, Multinomial performs slightly better than FLDA on this dataset, especially after removing outliers $\big($0.0837% (internal) and 0.18596 (external)% **vs.** 1.30% (internal) and 1.21% (external) $\big)$. However, both methods clearly perform better without outliers, which confirms that outliers have a negative impact on classification accuracy.

VII. Conclusion

These differences we found in the comparison section can be explained by the underlying assumptions of each method. FLDA relies on assumptions of approximate normality and equal covariance structures between classes. Since these assumptions are already violated in this dataset, the presence of outliers further distorts the estimates of the mean and covariance matrices, leading to a noticeable drop in performance. On the other hand, the Multinomial model does not rely on normality assumptions, making it more robust to deviations in the data distribution and the presence of outliers. This explains why its performance remains more stable across both cases.

VIII. Appendix

$$\text{Error Rate} = \frac{\text{Number of misclassified observations}}{\text{Total number of observations}} \tag{1}$$

Variable Name	Variable Type	Description	Unit of Measurement
Temperature	Interval	Measures the degree of heat or cold in the atmosphere	Degrees Celsius (°C)
Humidity	Interval	Represents the amount of moisture present in the air	Percentage (%)
Wind Speed	Interval	Indicates the speed of air movement in the atmosphere	km/h
Precipitation	Interval	Measures the amount of rainfall or other forms of water falling from the atmosphere	mm
Atmospheric Pressure	Interval	Represents the pressure exerted by the weight of air in the atmosphere	hPa
UV Index	Interval	Indicates the intensity of ultraviolet radiation from the sun	Index (unitless)
Visibility	Interval	Measures the distance at which objects can be clearly seen	km

Table I: Dataset Description

Actual / Predicted	**Cloudy**	**Rainy**	**Snowy**	**Sunny**	**Total**
Cloudy	2681	324	97	198	3300
Rainy	158	2796	252	94	3300
Snowy	98	39	3049	114	3300
Sunny	214	233	83	2770	3300
Total	3151	3392	3481	3176	13200

Table II: Fisher Linear Discriminant Confusion Matrix

Obs	**Cloudy**	**Rainy**	**Snowy**	**Sunny**
1	0.060294	0.591593	0.347060	0.001054
2	0.737387	0.079452	0.000815	0.182346
3	0.424042	0.020034	0.000381	0.555543
4	0.147175	0.604288	0.010955	0.237582
5	0.101013	0.855997	0.037062	0.002328

Table III: Posterior Probabilities for First 5 Observations

Class	**Correct Classification**	**Incorrect Classification**	**Total**
Snowy	3076	224	3300
Sunny	3127	173	3300
Total	6203	397	6600

Table IV: FLDA2 Classification Results

Actual / Predicted	**Cloudy**	**Rainy**	**Snowy**	**Sunny**
Cloudy	525	66	24	36
Rainy	32	556	58	24
Snowy	22	6	600	19
Sunny	46	44	20	562

Table V: Fisher Linear Discriminant Confusion Matrix (External Validation)

	Cloudy	Rainy	Snowy	Sunny
Cloudy	2565	12	0	3
Rainy	7	2874	85	0
Snowy	0	0	2671	0
Sunny	33	0	0	2502

Table VI: Confusion Matrix (Internal Validation - FLDA)

Obs	Cloudy	Rainy	Snowy	Sunny
1	0.9997651	2.60e-05	5.69e-12	2.08e-04
2	0.9999355	8.18e-06	1.55e-09	5.63e-05
3	0.9999945	4.66e-06	6.43e-07	2.05e-07
4	0.9943655	2.57e-06	3.98e-13	5.63e-03
5	0.9999994	1.39e-07	1.83e-13	4.71e-07

Table VII: Posterior Probabilities (First 5 Observations)

	Cloudy	Rainy	Snowy	Sunny
Cloudy	489	2	0	2
Rainy	2	584	17	0
Snowy	0	0	546	0
Sunny	3	0	0	506

Table VIII: Confusion Matrix (External Validation - FLDA)

Obs	Cloudy	Rainy	Snowy	Sunny
4	0.9945844	2.76e-06	3.88e-13	5.41e-03
6	0.9998729	7.62e-07	7.39e-14	1.26e-04
10	0.9999995	1.04e-08	2.51e-11	4.84e-07
12	0.9999948	1.70e-07	2.55e-12	5.01e-06
14	0.9998583	3.32e-08	2.66e-14	1.41e-04

Table IX: Posterior Probabilities (First 5 Observations - External Validation)

Class	Correct Classification	Incorrect Classification	Total
1	2534	1	2535
2	2671	0	2671
Total	5205	1	5206

Table X: FLDA2 Classification Results (p=2, k=2)

Actual / Predicted	**Cloudy**	**Rainy**	**Snowy**	**Sunny**
Cloudy	3300	0	0	0
Rainy	0	3300	0	0
Snowy	0	0	3300	0
Sunny	0	0	0	3300

Table XI: Perfect Classification Results (Multinomial Internal Validation

Actual / Predicted	**Cloudy**	**Rainy**	**Snowy**	**Sunny**
Cloudy	526	57	26	42
Rainy	28	585	32	25
Snowy	20	5	603	19
Sunny	42	39	19	572

Table XII: Multinomial Regression Confusion Matrix (External Validation)

	Cloudy	Rainy	Snowy	Sunny
Cloudy	2574	6	0	0
Rainy	3	2963	0	0
Snowy	0	0	2671	0
Sunny	0	0	0	2535

Table XIII: Multinomial Confusion Matrix (Internal Validation)

	Cloudy	Rainy	Snowy	Sunny
Cloudy	490	2	0	1
Rainy	1	602	0	0
Snowy	0	0	546	0
Sunny	0	0	0	509

Table XIV: Multinomial Confusion Matrix (External Validation)

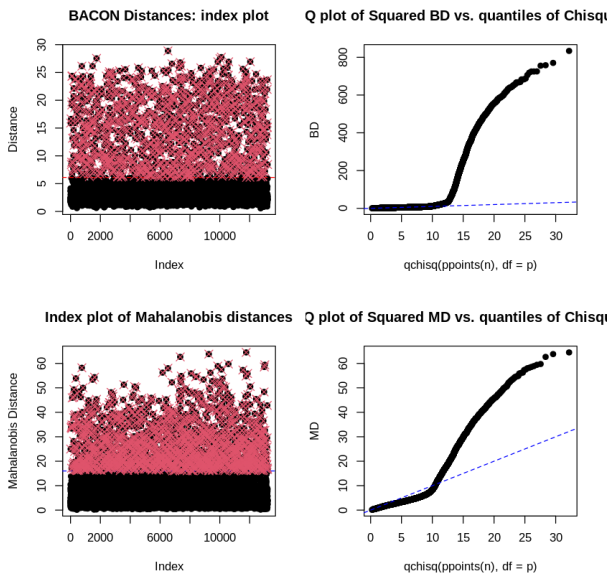


Figure I

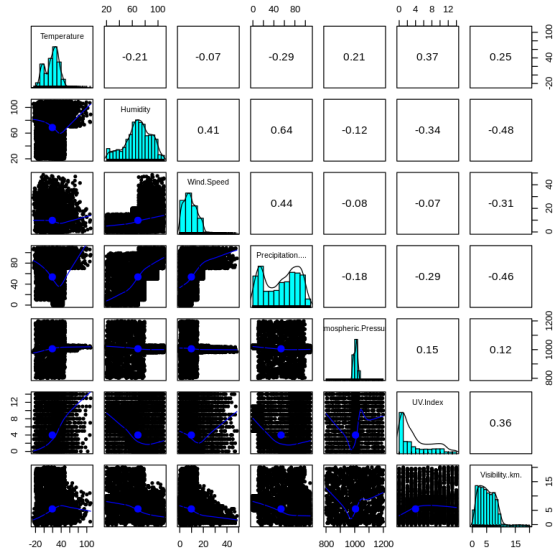


Figure II

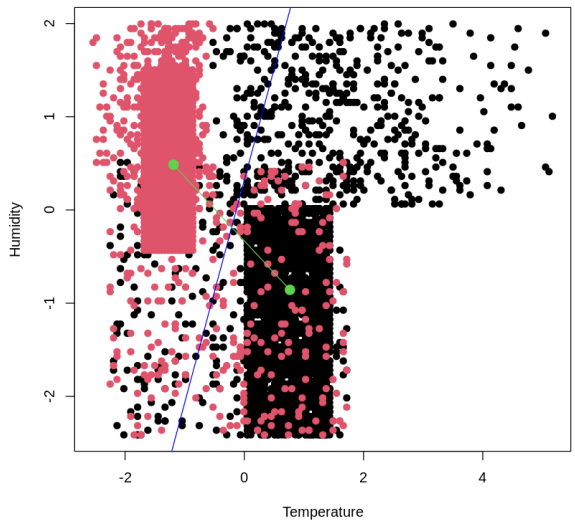


Figure III

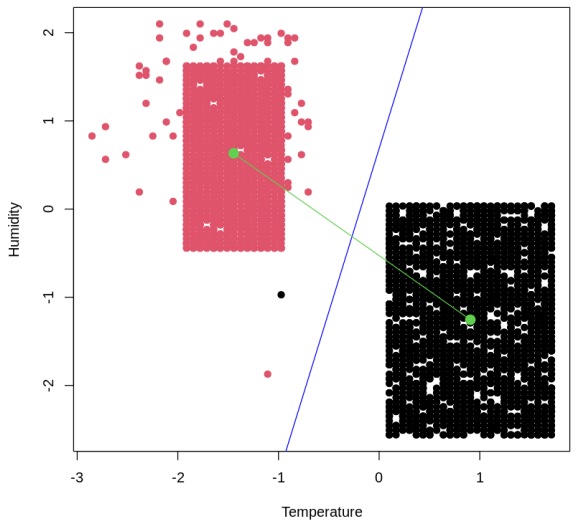
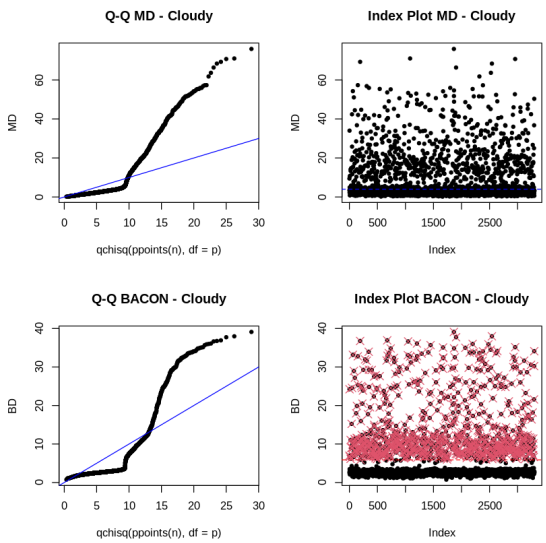
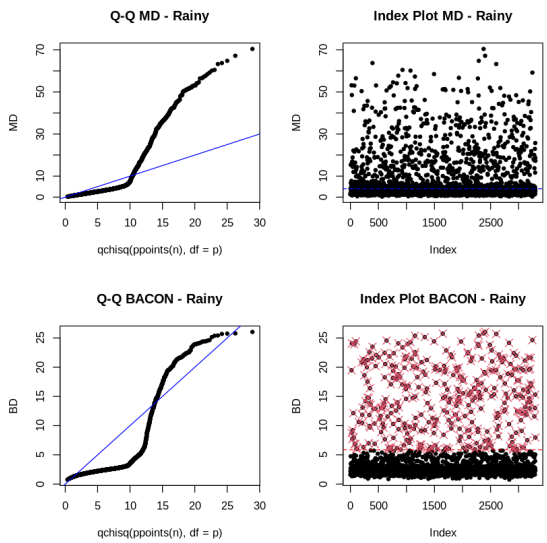


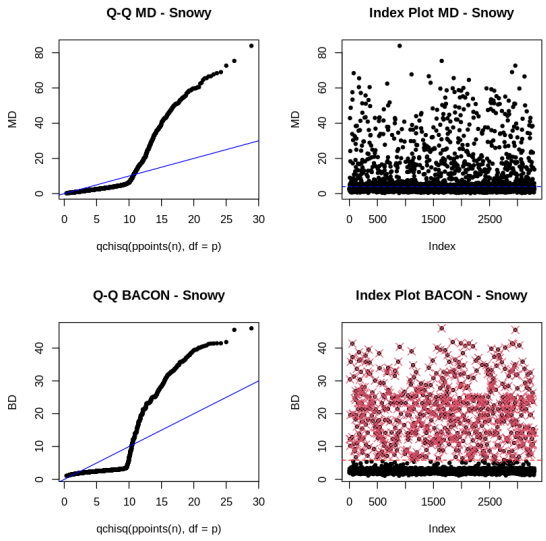
Figure IV



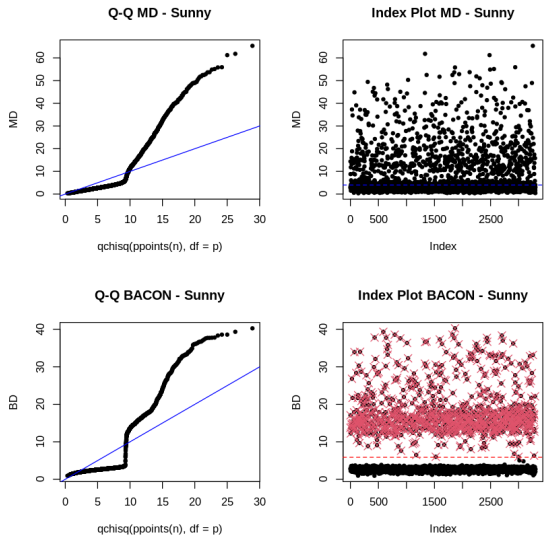
(a) Class Cloudy



(b) Class Rainy



(c) Class Cloudy



(d) Class Sunny

Figure V: Weather Classification Plots

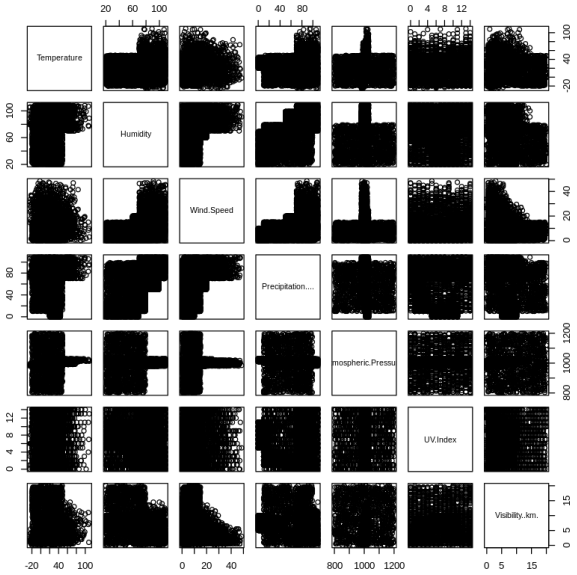


Figure VI: Pairwise Scatterplot